(Please write your Enrollment Number) Enrollment No._____

END TERM EXAMINATION

(CBCS)(SUBJECTIVE TYPE)

B Tech IT/MAE/DMAM, 2nd Semester

(AUGUST, 2022)

Subject Code: BAS 102	Subject: Applied Mathematics II
Time:3 Hours	Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(5*4 =20)**
	(a) Solve the following initial value problem using operator method : $y'' + y' - 2y = 0, y(0) = 0, y'(0) = 3$	
	(b) Find the conjugate harmonic function of $u(x,y) = e^x cosy$ and the corresponding analytic function.	
	(c) Find the Laplace transform of the function $g(t) = \begin{cases} 0 & , \quad 0 < t < 5 \\ t-3 & , \quad t > 5 \end{cases}$	
	(d) Evaluate $I = \int_0^1 \left(te^{-t^2} + 2i/\sqrt{t} \right) dt$	
	UNIT-I	
Q2	(a) Solve $y'' + 5y' + 6y = e^{-2x} sin2x$ using operator method (b) Using the method of undetermined coefficients, find the general solution of the differential equation $y'' - 3y' - 10y = 1 + x^2$	**(4+6=10)**
Q3	Solve the following simultaneous equations using operator method: $y'' + y' - 2y = \sin t \ , \ x' + x - 3y = 0$	**(10)**
	UNIT-II	
Q4	(a) Solve the following initial value problem using Laplace transform $y'' + 3y' + 2y = e^{-t}, y(0) = 0, y'(0) = -1$ (b) Find $L^{-1}\left\{\frac{4s+5}{(s-1)^2(s+2)}\right\}$	**(5+5=10)**
Q5	State and prove Convolution Theorem and hence solve $f(t) = \cos t + e^{-t} \int_0^t f(t-\tau) e^{\tau} d\tau$	**(10)**
	UNIT-III	
Q6	(a) If $f(z) = u + iv$ is an analytic function of $z = x + iy$ and $u - v = e^{-x}[(x-y)\sin y - (x+y)\cos y]$ Then find u, v and the analytic function $f(z)$ (b) Find and draw the image of the strip $2 < y < 4$ under the mapping $w = \frac{1}{z}$	**(7+3=10)**
Q7	(a) Show that the function $f(z) = \sqrt{\lvert xy \rvert}$ is not analytic at the origin, although the C-R equations are satisfied at origin. (b) Show that the function $f(z)$ is not continuous at $z = 0$, where $f(z) = \begin{cases} \frac{Re(z^2)}{\lvert z \rvert^2} & , \ z \neq 0 \\ 0 & , \ z = 0 \end{cases}$	**(6+4=10)**

	UNIT-IV	
Q8	(a) Evaluate the integral $I = \int_0^{2\pi} \frac{d\theta}{1-2a\cos\theta+a^2}$, $\lvert a\rvert < 1$ Where 'a' is a complex constant (b) Determine and classify all the singularities of the function $e^{\frac{z}{z-2}}$	(7+3=10)
Q9	(a) Evaluate $I = \oint \frac{z^2+4}{z^3+2z^2+2z}\, dz$ over $\lvert z-1\rvert = 5$ (b) Find the Laurent series expansion of the function $f(z) = \frac{7z^2+9z-18}{z^3-9z}$ about its singular points for the region $3 < \lvert z-3\rvert < 6$	(6+4=10)